SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

MURIEL · registered flood-depth mask

Synthetic MURIEL test district · not Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

- Inside registered MURIEL depth mask
- Outside registered MURIEL depth mask
- Unknown source depth / NoData

READ THIS MAP

Area >2000 m²; max depth
>0.3 m. Receptor <1 m;
municipal non-road land <50 m;
usable destination ≤100 m.
Protect overlap ≥70%.

Grey: buildings / receptors.
Pale blue: flood footprint.
Beige: roads. Pale green: city land.

Schwerdorf (2022); thesis §§4.4.2,
5.5.3. Reasons and evidence are
stored with every feature.

Footprint: depth_mask

Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: Input polygon partition; source resolution applies
Vertical datum: SYNTHETIC_DHHN2016
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_MURIEL_SRI5_V1
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Schwerdorf (2022), thesis §§4.4.2 / 5.5.3. Local planning screen; no infiltration or construction approval. SYNTHETIC DATA.

SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

MURIEL · flood hotspot screen

Synthetic MURIEL test district · not Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

- Area and maximum-depth criteria met
- Below local hotspot screen

READ THIS MAP

Area >2000 m²; max depth >0.3 m. Receptor <1 m; municipal non-road land <50 m; usable destination ≤100 m. Protect overlap ≥70%.

Grey: buildings / receptors.
Pale blue: flood footprint.
Beige: roads. Pale green: city land.

Schwerdorf (2022); thesis §§4.4.2, 5.5.3. Reasons and evidence are stored with every feature.

Footprint: depth_mask

Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: Input polygon partition; source resolution applies
Vertical datum: SYNTHETIC_DHHN2016
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_MURIEL_SRI5_V1
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Schwerdorf (2022), thesis §§4.4.2 / 5.5.3. Local planning screen; no infiltration or construction approval. SYNTHETIC DATA.

SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

MURIEL · protect, investigate or adapt

Synthetic MURIEL test district · not Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

- Below local hotspot screen
- Protect existing retention
- Retrofit · route evidence supplied
- Investigate topography / flow route
- No usable destination · alternatives
- Outside local adjacency / land screen
- Assessed route incompatible

READ THIS MAP

Area >2000 m²; max depth
>0.3 m. Receptor <1 m;
municipal non-road land <50 m;
usable destination ≤100 m.
Protect overlap ≥70%.

Grey: buildings / receptors.
Pale blue: flood footprint.
Beige: roads. Pale green: city land.

Schwerdorf (2022); thesis §§4.4.2,
5.5.3. Reasons and evidence are
stored with every feature.

Footprint: depth_mask

Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: Input polygon partition; source resolution applies
Vertical datum: SYNTHETIC_DHHN2016
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_MURIEL_SRI5_V1
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Schwerdorf (2022), thesis §§4.4.2 / 5.5.3. Local planning screen; no infiltration or construction approval. SYNTHETIC DATA.

SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

MURIEL · usable municipal destinations

Synthetic MURIEL test district · not Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

Usable municipal non-road space

READ THIS MAP

Area >2000 m²; max depth >0.3 m. Receptor <1 m; municipal non-road land <50 m; usable destination ≤100 m. Protect overlap ≥70%.

Grey: buildings / receptors.
Pale blue: flood footprint.
Beige: roads. Pale green: city land.

Schwerdorf (2022); thesis §§4.4.2, 5.5.3. Reasons and evidence are stored with every feature.

Footprint: depth_mask

Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: Input polygon partition; source resolution applies
Vertical datum: SYNTHETIC_DHHN2016
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_MURIEL_SRI5_V1
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Schwerdorf (2022), thesis §§4.4.2 / 5.5.3. Local planning screen; no infiltration or construction approval. SYNTHETIC DATA.

SYNTHETIC DEMONSTRATION · NOT A KÖLN-KALK RESULT

MURIEL · routing evidence and gaps

Synthetic MURIEL test district · not Köln-Kalk / SYNTHETIC_SRI5



Map key

Classification

- Route evidence supplied
- Incompatible assessed route
- Proximity only · no flow proof

READ THIS MAP

Area >2000 m²; max depth >0.3 m. Receptor <1 m; municipal non-road land <50 m; usable destination ≤100 m. Protect overlap ≥70%.

Grey: buildings / receptors. Pale blue: flood footprint. Beige: roads. Pale green: city land.

Schwerdorf (2022); thesis §§4.4.2, 5.5.3. Reasons and evidence are stored with every feature.

Footprint: depth_mask

Dashed lines show proximity or incomplete evidence; no flow proof.

Scenario: SYNTHETIC_SRI5
Need threshold: ≥10 cm
Horizontal CRS: EPSG:25832
Grid: Input polygon partition; source resolution applies
Vertical datum: SYNTHETIC_DHHN2016
Rule profile: koeln_thesis_provisional_v1
Run: DEMO_MURIEL_SRI5_V1
Source details: GeoPackage registers
Identify a feature for all rule decisions.

Schwerdorf (2022), thesis §§4.4.2 / 5.5.3. Local planning screen; no infiltration or construction approval. SYNTHETIC DATA.